Jagan Kumar Swain

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate (CGPA 9.1) spanning VLSI/mixed-signal design, embedded hardware, drone systems, and deep learning. First author of two research manuscripts: a 15-architecture retinal disease classification benchmark (IISER Berhampur) and an open-source 10T SRAM compute-in-memory macro in SkyWater 130nm. Micro Drone intern at PE Geometry. Best Innovation Award (Chakravyuh 1.0) and Rs. 50,000 Defence R&D Grant for GuardX Sentinel.

EDUCATION

B.Tech, Electronics and Communication Engineering 2024 – 2028
NIST University, Berhampur, Odisha *CGPA: 9.1*

BS, Data Science and Applications 2025 – 2028
Indian Institute of Technology (IIT) Madras *CGPA: 8.1*

EXPERIENCE

Research Intern – Deep Learning for Medical Imaging Jun 2026 – Present
IISER Berhampur, Odisha (Dept. of Computer Science) *Supervisor: Dr. Anabik Pal*

- Benchmarked 15 deep learning architectures (CNN, attention, ViT, Swin, MaxViT, hybrid) for multi-label retinal disease classification on RFMiD (3,200 fundus images, 45 disease labels) using 5-fold stratified cross-validation and an identical PyTorch/timm training protocol.
- Best single model MaxViT-Tiny reached RIADD 0.710 $\pm$ 0.015; a 7-architecture, 5-fold probability-averaging ensemble reached RIADD 0.756, mean AUC 0.933, and mAP 0.543 on the 640-image hold-out (+0.046 RIADD, +0.082 mAP over the best single model).
- Ran ablations on 4 loss functions and 3 augmentation policies: medical augmentation added 0.02–0.04 RIADD over baseline, and ImageNet pretraining lifted ViT-Small RIADD from 0.480 to 0.690. Applied Wilcoxon signed-rank tests across folds.
- Proposed the follow-on project *Hybrid Deep Learning Framework for Multi-Retinal Disease Localization, Segmentation and Classification*, approved by the supervisor; set up a private GitHub repository with a pull-request workflow for the team.

Engineering Intern – Micro Drone Project 05 Jun 2026 – 31 Jul 2026
PE Geometry India Pvt Ltd, Bengaluru, Karnataka *Geometry Assurance Engineering*

- Contributed to the development of a micro drone across its mechanical and electronics aspects.
- Applied the product design process to integrate mechanical and electronic subsystems for optimal product functionality.

RESEARCH MANUSCRIPTS

- J. K. Swain, S. R. Prusty, S. Mallik, A. Pal, "Systematic Benchmarking of Deep Learning Architectures for Multi-Label Retinal Disease Classification on the RFMiD Dataset." Manuscript in preparation.
- J. K. Swain, "Design, Mixed-Signal Modeling, and Statistical Characterization of an Open-Source 10T SRAM Compute-In-Memory Macro for Binary/Ternary Neural Networks in SkyWater 130nm CMOS." Manuscript in preparation (IEEE TCAS-I format).

TECHNICAL SKILLS

Programming Languages: C, Python, Java, JavaScript, HTML, CSS, Verilog
VLSI & Mixed-Signal: SkyWater SKY130 PDK, SPICE and Monte Carlo simulation, Yosys synthesis, static timing analysis, Cocotb, Icarus Verilog, Vivado, SRAM and compute-in-memory design, FPGA
AI / Machine Learning: PyTorch, timm, Deep Learning, CNN, LSTM, Vision Transformers, Quantization-Aware Training, Computer Vision, Medical Image Analysis, NLP
Embedded & Hardware: Arduino, Raspberry Pi, IoT, Digital Signal Processing, Drone Systems, EDA Tools
Web & Backend: Node.js, Express, MongoDB, Flask, REST APIs, JWT Authentication
Tools & Platforms: Linux, Git, GitHub, MATLAB, Google Colab, Tableau, AWS, Google Cloud Platform, Vertex AI

PROJECTS

Open-Source 10T SRAM Compute-In-Memory Macro (SkyWater 130nm)
VLSI, Mixed-Signal Design, Binary/Ternary Neural Networks

- Designed a 32$\times$32 differential 10T SRAM compute-in-memory macro with a decoupled read stack: 790 mV static noise margin at 1.8 V and zero compute read-disturb (ΔSNM = 0 mV).
- Built a 16-tap digitally controlled delay line (0.8–2.5 ns) and a 4-bit Flash ADC with StrongARM latches and a majority-voting Wallace tree decoder; a 16-row sub-slice datapath removed ADC saturation clipping (INL max 0.36 LSB).
- Synthesized the digital periphery with Yosys on sky130_fd_sc_hd, closing timing at 200 MHz (Fmax 648.9 MHz, 4,774 cells).
- Verified with a Cocotb/Icarus closed-loop co-simulation using a Python surrogate model under Pelgrom mismatch and comparator offset: 99.94% of 1,600 dot products within $\pm$1 LSB.
- Emulated a quantization-aware LeNet-5 on MNIST: 97.62% accuracy (0.24% below software) and at least 95.95% across a 15-point PVT grid.

GuardX Sentinel – Drone-Based Human Activity Detection System
Computer Vision, IoT, Embedded Systems

- Developed an indigenous drone-integrated surveillance system for real-time human activity detection and threat identification in defense scenarios.
- Integrated Arduino Nicla Vision, thermal and regular cameras, and onboard sensors for low-latency edge processing under constrained compute.
- Won **Best Innovation Award** at Chakravyuh 1.0 and a **Rs. 50,000 Defence R&D Grant** at the International Conference on Air Defence and Security.

SegroBin – Smart Waste Management System (Smart India Hackathon 2025)
IoT, Embedded Systems, Application Development

- Designed a smart waste segregation system for Government of Odisha problem statement SIH25046; Top 5 finalist for the problem statement.
- Built an IoT-enabled smart dustbin with a companion app for real-time tracking and a Green Credits incentive for recyclable segregation.

DeepBeat – ECG Arrhythmia Classification System
Deep Learning, Medical Time-Series Analysis

- Built a hybrid CNN–LSTM in PyTorch on the MIT-BIH dataset (87,000+ ECG samples): 96% overall accuracy with high recall on critical ventricular and supraventricular classes.

Formula 1 Race Winner Prediction System
Machine Learning, Time-Series Analysis

- Engineered driver-form, season-momentum, circuit-affinity, and grid-position features from Kaggle data; compared Logistic Regression, Random Forest, Gradient Boosting, and Neural Network models, reaching 95.6% cross-validated accuracy with time-aware validation.

16-Operation ALU Design using Verilog HDL
Verilog, Digital Design, VLSI

- Designed a 16-operation ALU (arithmetic, logical, shift, comparison, Carry and Zero flags) and verified function and timing through waveform simulation in Vivado 2025.2.

AI Career Agent – IndiaAI Impact GenAI Hackathon (IISc Bengaluru, MeitY)
AI Agents, Full-Stack Development

- Built a career guidance agent (skill extraction, job-role matching, learning paths) with a Flask backend, AI reasoning engine, and SPA frontend using IBM open-source agentic workflows.

Additional Projects
Full-Stack, Embedded Systems

- Full-Stack Authentication System: Node.js, Express, MongoDB with JWT, bcrypt hashing, and input validation.
- Smart Blind Stick: Arduino obstacle detection with real-time alerts. Arduino 3D Scanner: stepper-motor turntable synchronized with camera capture.

ACHIEVEMENTS

- **Rs. 50,000 Defence R&D Grant**, Army AD College and NIST University, International Conference on Air Defence and Security (GuardX Sentinel).
- **Best Innovation Award**, Chakravyuh 1.0 (GuardX Sentinel).
- **Grand Finalist**, Smart India Hackathon 2025, Hardware Edition (Team CtrlOhm).
- **1st Place**, Web Development Competition, Festronix 2K26, GIFT Autonomous, Bhubaneswar.
- **Winner**, Code Crusade 4.0 (1st and 2nd Year category), Club Excel, NIST University.

CERTIFICATIONS AND TRAINING

- **Build and Deploy Machine Learning Solutions on Vertex AI**, Google Cloud Skill Badge.
- **Data Analytics Job Simulation**, Deloitte Australia (Forage): Tableau dashboards and Excel insights.
- **Internet of Things Internship**, LaunchED Global; **IoT and Mechatronics Internship**, NIST University: Arduino, sensors, actuators, Blynk, OTA updates, cloud IoT.
- **OOSC 3.0 Open-Source Software Conference**, IIT Kanpur; **DevOps Webinar**, Malla Reddy Engineering College and GDG; **Google and GDG Tech Event** (Vertex AI, Gemini).
- **Workshops**, NIST University: SensoTech (IoT), Blender and AR, AI Cloudwave (Nov 2024).